IT Support Cost Optimization Using Data Analysis

CountryEdu / Abhishek & Company — Data Analyst Case Study

Synthetic demonstration project built from the supplied problem statement. The actual company dataset was not supplied, so all numerical outputs below are from a synthetic 200,000-ticket sample.

1. Executive Summary

The analysis framework calculates ticket-level support cost, identifies cost drivers, evaluates escalation behavior and customer satisfaction, and supports engineer-allocation decisions. In the demonstration sample, modeled total support cost is 12,683,069 currency units, average cost per ticket is 63.73, average resolution time is 1.89 hours, escalation rate is 12.86%, and average CSAT is 4.56. These figures are illustrative only and must be replaced with real company data before making business claims.

2. KPI Framework

KPI	Definition	Target
Total Support Cost	Σ resolution hours × engineer hourly cost	Reduce 20%
Cost / Ticket	Total cost ÷ tickets	Downward
Escalation Rate	Escalated tickets ÷ tickets	Downward
Average CSAT	Mean customer satisfaction score	> 4.2
Avg Resolution Time	Mean resolution duration	Downward
Engineer Utilization	Productive hours ÷ available hours	Balanced

3. Cost Driver Results

The following synthetic results demonstrate how the final analysis should be presented. Categories should be ranked by total cost rather than ticket volume alone.

Category	Tickets	Total Cost	Cost Share	Escalation	CSAT
Network	40,048	3,065,163	24.2%	18.8%	4.52
Hardware	25,903	3,017,090	23.8%	20.0%	4.49
Software	52,058	2,820,903	22.2%	11.2%	4.57
Database	8,074	996,364	7.9%	25.7%	4.47
Security	9,959	936,755	7.4%	24.4%	4.49
Access	35,924	911,273	7.2%	3.4%	4.63
Email	20,140	591,704	4.7%	4.2%	4.62
Other	7,894	343,818	2.7%	7.7%	4.59

4. Missing Data Strategy

The case permits up to 20% missing data. The project does not drop incomplete records wholesale. Numerical fields are imputed using context-aware medians, categorical client type is assigned an explicit Unknown category, and production pipelines should additionally retain missingness flags. Missingness should be tested for systematic bias before modeling.

5. Escalation Prediction

A logistic-regression baseline was trained to predict whether a ticket will escalate using issue category, priority, client type, resolution time and engineer cost. On the synthetic holdout set, ROC-AUC was 0.720. The model is deliberately simple and fast so it can serve as a production baseline. A later iteration could benchmark gradient boosting and calibrated probabilities.

6. Real-Time Architecture

At 50,000 events per second, a streaming architecture is appropriate. Events can enter Kafka or an equivalent event bus, pass through Spark/Flink-style stream processing, receive pre-computed features, and call a lightweight model service. The model should return an escalation-risk score well below the 5-second business SLA, with the remaining latency budget reserved for ingestion and downstream ticket routing.

7. Cost Reduction Plan

Workstream	Illustrative target contribution	Action
Automation	8 pp	Self-service, knowledge base, automated workflows
Escalation reduction	5 pp	Better routing, first-line playbooks
Engineer reallocation	4 pp	Balance workload and expertise
Resolution improvement	3 pp	Reduce bottlenecks in high-cost categories
Total	20 pp	Validate against actual baseline

8. Final Recommendation

Management should not pursue a blanket 20% reduction in engineer hours. The safer strategy is targeted optimization: automate repetitive tickets, reduce unnecessary escalations, improve routing and rebalance engineer capacity while protecting CSAT. The dashboard should report cost and customer outcomes together so savings are not achieved at the expense of service quality.

Important: Replace all synthetic numerical findings with results calculated from the employer's actual dataset if it is provided.